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Medical Chatbot using Gamma LLMV2 and Comparison Using BERT Models

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***Abstract*—The study introduces a sophisticated medical chatbot that uses vector-based retrieval and Meta's LLaMA 2 model to provide accurate, context-aware recommendations on symptoms, drugs, and diets. The system incorporates Pinecone for vector embeddings, LangChain for interactions, and Python for logic. "The GALE Encyclopedia of Medicine," a 637-page medical dataset, is broken up into text segments for effective semantic search.Flask is used for web infrastructure and Streamlit for interaction with the chatbot's front end. User queries use LLaMA 2 to deliver answers, create embeddings, and do vector searches. While handling edge circumstances and data quality provide issues, evaluation emphasizes accuracy, timeliness, and engagement. Real-time updates and sophisticated fine-tuning are examples of upcoming enhancements.In terms of language interpretation, generation, and reasoning, Gamma LLM v2 performs better than proprietary models when compared to other models. This enables fine-tuning on bespoke datasets and lessens the need for APIs. It outperforms RoBERTa (0.77), MedBERT (0.95), and BERT (0.86) in medical question-answering tasks and is available in 7B, 13B, and 70B parameters.**

***Index Terms*—Gamma LLM-V2,Chat bot, BioBERT, MedBERT, Langchain.**

I. Introduction

Intelligent healthcare systems are now possible because to developments in AI and NLP. Using Gamma LLMV2, this study creates a medical chatbot and evaluates it against BERT variants. The chatbot uses Meta's LLaMA 2 to provide precise, context-aware answers when assisting with symptoms, prescriptions, and diet. The system incorporates Python for backend functionality, Pinecone for vector embeddings, and LangChain for conversational interactions. For effective semantic search, a 637-page medical dataset is preprocessed into text segments and embedded. Built with Flask and Streamlit, the front end offers an easy-to-use interface. To produce answers, LLaMA 2 processes user inputs once they have been integrated and searched in Pinecone. The system's potential to improve patient involvement is highlighted by evaluations based on accuracy, reaction time, and user happiness. Managing medical edge cases, maintaining data quality, and integrating real-time medical updates are among the difficulties.

II. Literature Review

The creation of medical chatbots is the main topic of the literature study, which examines the advancement of AI and NLP technologies in healthcare. It reviews earlier research on big language models, vector-based retrieval systems, and medical applications. This section outlines the main developments and difficulties in developing precise, context-aware chatbots for the medical field.

The study compares Gamma LLM v2's effectiveness with other models, such as BERT versions like MedBERT and RoBERTa, and focuses on examining how it may improve the performance of medical chatbots, specifically in providing precise and context-aware advice. Identifying and resolving the main obstacles to the deployment of large language model-based (LLM-based) medical chatbots is another goal of the project. The study objectives include creating and deploying a medical chatbot using Gamma LLM v2 to increase accuracy and user happiness in order to accomplish these aims. In order to provide a thorough evaluation of the chatbot's efficacy in responding to medical inquiries, it also aims to assess its performance against BERT, MedBERT, and RoBERTa models using medical datasets.

The article by Akilesh S. etal shows the integration of large language models (LLMs) into healthcare has been widely explored due to their ability to process complex medical queries and provide accurate, context-aware information. Studies on GPT-3.5 and similar models highlight their capacity to understand and generate human-like text, making them suitable for medical applications such as diagnosis support, patient interaction, and personalized recommendations. Research emphasizes the role of natural language processing (NLP) in enhancing user engagement and facilitating seamless communication between patients and healthcare systems.[1]

The work by Mengfei Li atal. has shown the application of artificial intelligence (AI) in medicine has gained significant attention, with language models like ChatGPT being utilized for tasks such as text analysis ,summarization, and data mining.Machine learning techniques, including supervised and semi-supervised learning, have been widely used for entity recognition in domains like prescription analysis. These approaches have demonstrated effectiveness in identifying and categorizing medical terms, especially in historical and non-standardized datasets. Studies have also revealed the potential of AI in analyzing ancient medical practices, offering a historical perspective on clinical treatments and drug formulations.[2].

The research work done by Mohammad Abu Tareq Rony ,etal. demonstrated the recent advancements in generative pre-trained transformers, particularly GPT models, have opened new avenues for medical text processing.Several datasets, such as the Illness-dataset, Symptom2Disease, and clinical trial protocols, have been employed to evaluate the performance of AI models in medical text classification. These datasets provide diverse challenges, including short-form text from social media, structured clinical data, and symptom-to-disease mapping. ChatGPT and its enhanced adaptations, such as MediGPT, have demonstrated substantial improvements in accuracy and robustness across these datasets, outperforming many conventional models and reducing the dependency on domain-specific customization.[3].

Recent studies by Satya Prakas.etal highlight the efficacy of LSTM variants—such as Bi-Directional LSTM, Stacked LSTM, and hybrid models like CNN-LSTM and GRU-LSTM—in time series forecasting. These models incorporate advanced mechanisms for capturing long-term dependencies and integrating external features, making them well-suited for dynamic datasets like COVID-19 case counts. Additionally, attention mechanisms have been explored to enhance interpretability and focus on critical time points in data sequences.[4].

Chen, Kou, Bai, and Tong (2023) address the challenge of overfitting in fine-tuned BERT models, particularly when working with small datasets. They identify that irrelevant or misleading words in sentences can significantly degrade model performance. Their approach integrates SSA into BERT through two methods and combines them in a hybrid framework to leverage their strengths. Empirical results on public datasets demonstrate that SSA-enhanced BERT models achieve notable improvements in performance, highlighting the technique's effectiveness in overcoming fine-tuning limitations on small datasets.[5].

Vasantharajan, Tun, Thi-Nga, Jain, Rong, and Siong (2023) present MedBERT, a pre-trained transformer-based model specifically designed for biomedical named entity recognition (NER). Trained on 57.46M tokens from diverse biomedical data sources such as N2C2, BioNLP, CRAFT, and Wikipedia, MedBERT addresses the unique language patterns in biomedical text. Comparative evaluations against four existing pre-trained models on ten datasets from BioNLP and CRAFT shared tasks demonstrate MedBERT's superior performance. The model achieved state-of-the-art results in nine datasets, excelling in predicting entities like Protein, Gene, and Chemical, with an average F1-micro score of 84.04 . Notably, MedBERT outperformed BioBERT .[6].

Srivastava and Singh (2023) developed an automated medical chatbot aimed at providing individualized diagnoses through conversational engagement with patients. The chatbot identifies symptoms from user inputs, achieving a precision of 65 percent and a recall of 65 percent for symptom recognition, and delivers an expected diagnosis with a precision of 71 percent. This demonstrates the feasibility of a conversational medical bot in performing basic symptom analysis and diagnosis, suggesting its potential to improve access to healthcare services. The findings indicate that further advancements in automated medical systems could enhance their role in healthcare, paving the way for more effective and scalable solutions.[7].

Zhao, Wang, Gu, Zhu, Mei, and Zhuang (2023) introduce ChatCAD+, a novel integration of Computer-Aided Diagnosis (CAD) with Large Language Models (LLMs) aimed at improving diagnostic processes and virtual consultations in healthcare. Additionally, the lack of deep medical expertise in current LLMs reduces their effectiveness as reliable virtual family doctors. To address these issues, ChatCAD+ is designed with two key modules: Reliable Report Generation, ChatCAD+ aims to replicate the expertise of human professionals, enhancing the consistency and reliability of both diagnostic reports and patient consultations. This approach represents a significant step toward making automated medical systems more universally applicable and trustworthy in clinical settings.[8].

By using methods like n-grams, TF-IDF, and cosine similarity for query responses, medical chatbots that use AI and NLP improve healthcare access (Lekha Athota, Vinod Kumar Shukla, Nitin Pandey, and Ajay Rana). The authors stress the value of effective database management for scalability and the function of fallback mechanisms managed by expert systems to enhance reliability. Accuracy and trust issues still exist despite the progress made, so algorithmic and expert integration improvements must be ongoing. Building on these approaches, this study suggests a healthcare solution that is easier to access.[10]

Using Convolutional Neural Networks (CNNs) for feature extraction and Long Short-Term Memory (LSTM) or Bidirectional LSTM (BiLSTM) for sequential decoding, Yeeshant Dahikar, Diptee Chikmurge, and Sharmila Kharat investigate sketch captioning. CNNs record both local and global sketch features, LSTMs model temporal relationships, and BiLSTMs use bidirectional analysis to improve context understanding. Their approach, which combines sequence generation and supervised learning with large paired sketch-caption datasets, provides better captioning than current methods.[11].

The characteristics of LangChain improve overall productivity and greatly simplify PDF querying. Semantic search, driven by the latest Transformer language models, represents a significant evolution in information retrieval systems. [12].

Frameworks like OpenAI's Chat-GPT (GPT3.5 Turbo) and Pinecone are used for generating responses. The outcomes demonstrate that the chatbot is capable of generating coherent responses that are closely aligned with the context of the PDF document.[13].

The FAQ section answers the frequently asked questions of the user. Generally, these are answered by humans but not a chatbot. Hence, the paper proposes a new chatbot which can also answer Frequently asked questions.[14].

Utilizing the web-based framework alongside open-source technologies such as Spring Boot, Spring Security, Apache MyBatis, Vue, and Element UI, along with database technologies such as MySql, MongoDB, and Redis, authors analyze the basic elements of medical device data dictionary system and incorporate layered architecture technology, design the system architecture, functional workflows and database model required for the dictionary system, and conduct professional testing and trial evaluation of the system.[15].

The basic idea is to construct a healthcare chatbot (MedBot) based on Artificial Intelligence and Natural Language Processing, which can identify the illness and provide necessary information about it prior to consulting or visiting a doctor, thereby making the MedBot more reachable and reducing healthcare costs. [16].

The support vector machine (SVM) method is used as an optimal classifier that learns the classification hyperplane in a space map that has the maximal distance (margin) to the training examples. The SVM will predict the suggested specialist based on the given symptom and comorbidities by the users. [17].

The effectiveness of medical chatbots rest on Natural language processing technique that helps users post their concerns about any disease and also their health. The user can ask any private question related to health care via chatbot without being actually present at the clinic or hospital.[18].

A WhatsApp chatbot is a program that can reply to messages automatically on WhatsApp. In the paper, to make the healthcare system more effective, and interactive a diagnostic chatbot is designed and developed using machine learning algorithms, to help public users in booking, cancelling, and rescheduling Doctor's appointments based on their timings.[19].

The research article paper introduces a Healthcare Chatbot powered by Artificial Intelligence (AI) to facilitate basic health-related interactions and provide preliminary medical advice based on user symptoms before consulting a doctor. The system utilizes Natural Language Processing (NLP) for communication, analyzing user input through techniques like TF-IDF, stemming, n-grams, and cosine similarity to calculate sentence similarity and rank responses.[20].

III. SYSTEM ARCHITECTURE

Building a medical AI chatbot with NLP, deep learning, and a structured medical knowledge base is broken down step-by-step in the diagram. It describes the flow of the system and the function of each part, as illustrated in Fig. 1 (Architecture Diagram).

- **Medical Book (Source of Knowledge)** :The GALE Encyclopedia of Medicine" in PDF format serves as the system's starting point. This book offers useful medical information on a wide range of illnesses, symptoms, diagnostics, therapies, and more. The PDF serves as the primary source from which the chatbot will learn everything. This is a crucial component of the system because the data taken from this book will be used to inform the chatbot's responses.
- **Extract Data/Content from the Book** :The machine then extracts information or content from the book after locating the pertinent PDF. During this extraction process, the PDF is divided into digestible text pieces that the system can process more readily. These sections provide a variety of information, including references to specialists or procedures, symptoms, treatments, and descriptions of medical disorders. All of the book's material is automatically extracted, guaranteeing that no important details are overlooked.
- **Text Chunks**: In essence, the text chunks derived from the medical book are condensed passages of text that relate to particular subjects. A text fragment might, for example, be a portion that discusses disease symptoms or the procedures for a certain diagnosis. The system can better control the information flow and match it to user queries by breaking the book up into manageable parts. When consumers pose particular health-related queries, this segmentation enables a more detailed and accurate response system.
- **Embeddings**: Following their creation, the text chunks must be formatted such that the machine learning model can comprehend them. Here's where embeddings are useful. Embeddings are high-dimensional numerical representations of words or phrases that maintain their semantic meaning. For instance, the terms "temperature" and "fever" can be next to one another in the embedding space, suggesting that they are connected. To make sure the system can understand user queries based on meaning rather than just keywords, this step is essential. The approach enables semantic understanding—the ability to identify context and connections between various medical terms—by embedding the text chunks.
- **Building the Semantic Index**: The creation of a semantic index comes next after the embeddings are in place. In essence, the semantic index is a methodical approach to arranging and storing the embeddings in a manner that facilitates quick and effective retrieval. The system can swiftly look through this index to locate pertinent information from the text chunks when a user asks a question. The system matches the user's query with the most appropriate response from the knowledge base thanks to the semantic index, which functions as a search engine that comprehends the meaning of words and phrases.

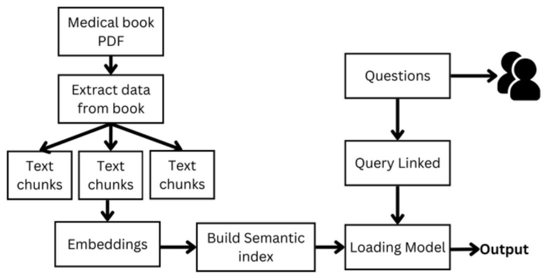

Fig. 1. Architecture Diagram

- **Question Input and Query Linking**: Users can enter their queries in natural language using the chatbot's user interface. A user might inquire, "What are the symptoms of COVID-19?" for instance. After that, the system will transform this query into one that can be handled. In the query linking step, the natural language query is compared to the pertinent text passages in the semantic index. Because the system must make sure it completely comprehends what the user is requesting, this phase is crucial. It accomplishes this by retrieving the most pertinent information by matching the semantics of the query with the indexed content.
- **Loading Model and Knowledge Base**: The system must produce an answer after linking the query. The structured knowledge base and the pre-trained model are used to do this. The pre-trained model is intended to comprehend user intent and react accordingly; it is probably based on NLP models like transformers or LSTM (Long Short-Term Memory). The data taken from the medical literature, which offers the factual information, makes up the knowledge base. The query is analyzed by the model, which then maps it to a pertinent piece of knowledge base data. The accuracy and contextual relevance of the responses are guaranteed by this dual method.
- **Output (Response to User)**: The system finally produces an output after processing the query. The user receives this output as a text response that either answers their query or gives them the information they asked for. The system might provide a list of typical COVID-19 symptoms taken from the medical book, for instance, if the user inquired about them. Users may easily acquire vital medical information thanks to the chatbot's ability to generate such responses, which guarantees a smooth and educational encounter.

Using AI methods like deep learning, natural language processing, and embeddings, the system offers rapid access to medical knowledge. From a structured knowledge base covering symptoms and treatments, it guarantees precise, context-aware responses. Contactless medical assistance is made possible by this architecture, which is essential in pandemic situations. The chatbot creates accurate real-time responses by converting medical PDFs into embeddings, which reduces healthcare burdens and enhances accessibility.

Parameter	Optimal Value
Temperature	0.8
Top-K Sampling	50
Model Size	13B
Number of Tokens	1024
Context Window	4096 tokens
Frequency Penalty	0.8
Presence Penalty	0.7

TABLE I
PARAMETERS FOR GAMMA LLMV2 MODEL

The parameters in above Table that have been adjusted for the GAMMA LLMv2 Model to improve its functionality and performance in medical chatbot applications are described in detail below:

1) Temperature: For replies that balance determinism and randomness, set to 0.8.
2) Top-K Sampling: Set to 50 to guarantee outputs that are relevant but varied.
3) Model Size: 13B was selected because of its balanced computing efficiency and precision.
4) Number of Tokens: 1024 is the limit for thorough yet succinct answers.
5) Context Window: For smooth multi-turn talks, all 4096 tokens were used.
6) Frequency Penalty: To discourage the use of tokens repeatedly, set to 0.8.
7) Presence Penalty: Reduce redundancy in generated responses by setting it to 0.7.

The overall system architecture uses text in form of chunks that are stored at the Pinecone database. The knowledge base is built on model and queries are retrieved by end user.

IV. RESULT AND ANALYSIS

The previously implemented chat bot was made to be compared with some other variations of BERT models like SBERT,RoBERTa,MedBERT,BERT. The pre-trained model was fine tuned using some steps and the final predictions were made for the test dataset. The result was calculated based on average score . The average score was defined as

The formula for calculating the average score using SBERT is as follows:

$$\text{average\_score\_sbert} = \frac{\sum_{i=1}^{n} \text{Rating}_i}{n} Eq.(1)$$

Explanation:

- $\sum_{i=1}^{n} \text{Rating}_i$: This represents the sum of all ratings in the dataset. Here, i ranges from 1 to n, where n is the total number of items in the dataset.
- n: This is the total number of items in the dataset.
- The formula divides the total sum of ratings by the number of items (n) to compute the average score.

Example:

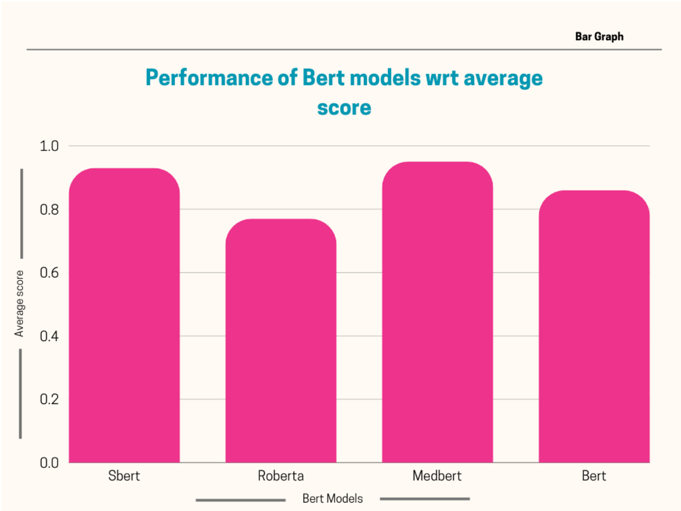

Fig. 2. Result Analysis

Given the dataset:

$$\text{evaluation\_data} = \begin{bmatrix} \{\text{Rating} : 4\}, \\ \{\text{Rating} : 5\}, \\ \{\text{Rating} : 3\} \end{bmatrix}$$

$$\sum_{i=1}^{3} \text{Rating}_i = 4 + 5 + 3 = 12$$

$$n = 3$$

$$\text{average\_score\_sbert} = \frac{12}{3} = 4.0$$

Thus, the average score is 4.0. from Eq.(1).

Similarly , the score for the rest of three models were calculated that were 0.77,0.95,0.86 for RoBERTa,MedBERT,BERT respectively. The model was fine tuned on Medical question-answering dataset.

The brief summary related to each of the models is discussed as below.

Sentence-BERT (SBERT)

Sentence-BERT (SBERT) is a modification of the BERT (Bidirectional Encoder Representations from Transformers) architecture that is designed to compute dense vector embeddings for sentences. It was introduced in the paper *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks* by Reimers and Gurevych [**?**]. SBERT offers a more efficient solution for tasks such as semantic similarity, clustering, and information retrieval compared to the original BERT model.

Key Features

- **Purpose:** SBERT generates semantically meaningful sentence embeddings that can be directly compared using vector operations such as cosine similarity.
- **Architecture:** It uses a Siamese or triplet network structure with shared weights based on a BERT encoder. This allows it to process sentence pairs independently.
- **Efficiency:** SBERT computes embeddings for sentences once, making tasks like pairwise comparisons or clustering significantly faster than using BERT, where each pair needs to be processed together.
- **Optimization:** It is fine-tuned using loss functions like cosine similarity loss or triplet loss to ensure that embeddings of semantically similar sentences are closer in the vector space.

Applications

SBERT can be used for a wide range of tasks, including:

- **Semantic Textual Similarity:** Measuring the semantic similarity between two sentences.
- **Clustering:** Grouping similar sentences or texts.
- **Information Retrieval:** Retrieving sentences or documents most similar to a given query.
- **Question Answering:** Finding the most relevant passages or sentences that answer a given question.

BERT (Bidirectional Encoder Representations from Transformers)

BERT, introduced by Devlin et al. , is a Transformer-based model designed for natural language understanding (NLU). It is pre-trained using a bidirectional Transformer architecture to understand the context of a word based on both its left and right surroundings in a sentence.

Key Features:

- **Pre-training Tasks:** BERT uses two tasks during pre-training:
  1) **Masked Language Modeling (MLM):** Predicting randomly masked words in the input text.
  2) **Next Sentence Prediction (NSP):** Predicting whether two sentences appear consecutively in the original text.
- **Fine-tuning:** BERT can be fine-tuned for downstream tasks like sentiment analysis, question answering, and named entity recognition.

Applications:

- Question answering (e.g., SQuAD).
- Sentiment analysis.
- Text classification and token classification tasks.

RoBERTa (A Robustly Optimized BERT Pretraining Approach)

RoBERTa, proposed by Liu et al. [**?**], is an improved variant of BERT. It addresses several limitations in BERT's pre-training process and achieves superior performance on many benchmarks.

Key Improvements Over BERT:

- Removal of the Next Sentence Prediction (NSP) task.
- Longer training on larger datasets.
- Dynamic masking during pre-training for more diverse masked tokens.
- Larger batch sizes and learning rates.

Applications:

- Fine-grained text classification.
- Language generation when paired with autoregressive decoders.

MedBERT (Medical BERT)

MedBERT is a specialized version of BERT tailored for healthcare and medical data processing. It is pre-trained on medical-specific datasets, such as clinical notes, electronic health records (EHRs), and biomedical literature.
Key Features:

- **Domain-Specific Pre-training:** MedBERT is trained using medical corpora to adapt the model to healthcare terminology and context.
- **Applications:** Commonly used for tasks like:
  * Predicting patient outcomes.
  * Diagnosis classification.
  * Medical named entity recognition (NER).
- **Variants:** Versions like ClinicalBERT and BioBERT specialize in subdomains of the medical field.

The results highlight performance of models . The overall score generated was greater than 80 percent.

V. Conclusion

Gamma LLM v2 improves transformer-based models with an emphasis on conversational AI and healthcare. Bidirectional modeling was introduced by BERT, but it had trouble with sentence-pair tasks. These problems are resolved by Gamma LLM v2, which still has bidirectional capabilities. With dynamic masking, bigger datasets, and better pretraining, RoBERTa enhanced NLP tasks—advancements integrated into Gamma LLM v2 for healthcare applications.When it came to sentence-pair tasks, SBERT performed exceptionally well, optimizing BERT for semantic similarity. This is expanded to include context-aware medical discussions in Gamma LLM v2. In contrast to MedBERT, which lacked real-time fluency, it blends domain-specific fine-tuning with natural dialogue. It guarantees precise, compassionate healthcare interactions by integrating with medical records and APIs.

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